Exercises

A12-M01

List all matches of P=aba in T=ababa.

A12-M02

Where does the empty pattern match in a text of length 4?

A12-M03

What is the output for text length 3 and nonempty pattern length 5?

A12-M04

Compute π for P=abababaca and give the fallback chain at i=5.

A12-M05

Build the DFA for P=aba over {a,b}.

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A12-M06

With d=10 and q=13, do 12 and 25 collide? Are they equal?

A12-M07

Compute length-2 window hashes of 1234 with d=10 and q=13.

A12-M08

Find all positions of P=aaa in T=aaaaa using KMP.

A12-M09

Explain why KMP fallbacks are linear.

A12-M10

When should Rabin–Karp verify characters? How should negative remainders be handled? For T=a^n and P=a^⌊n/2⌋, count the verification work. Does having no false collisions imply linear time?

A12-X01

For P=aba and Σ={a,b}, use the π recurrence to compute the two transitions from accepting state q=3. Verify them directly using σ(P⟨3⟩a) and σ(P⟨3⟩b), and explain how these states preserve overlap.

A12-X02

For P=abababaca, list all proper border lengths of P⟨5⟩=ababa and justify completeness using the π chain. If the next symbol is x, trace every fallback and final comparison.

A12-X03

Let q=11, P=12, T=122312 and H_b(xy)=bx+y mod 11. List the hash, true or false hit, and output for each window with b=10. Then choose b uniformly from F_{11} and find the expected number of false hits.

A12-X04

Distinct length-5 strings X,Y have symbol encodings injective into F_{11}; choose b uniformly from F_{11}. Prove a bound on their collision probability. For three nonmatching windows and z true matches, bound the expected verification work and state whether independent collisions are needed.

References and attribution

CLRS, Introduction to Algorithms, 3rd ed., chapter 32, pp. 990–1006. MIT 6.046J Fall 2004, Lecture 22.

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